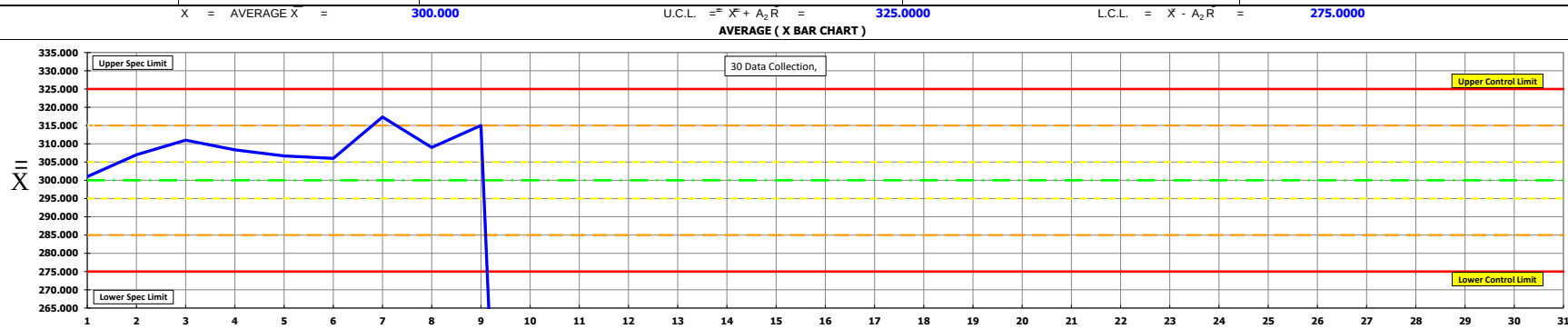


X - R CONTROL CHART

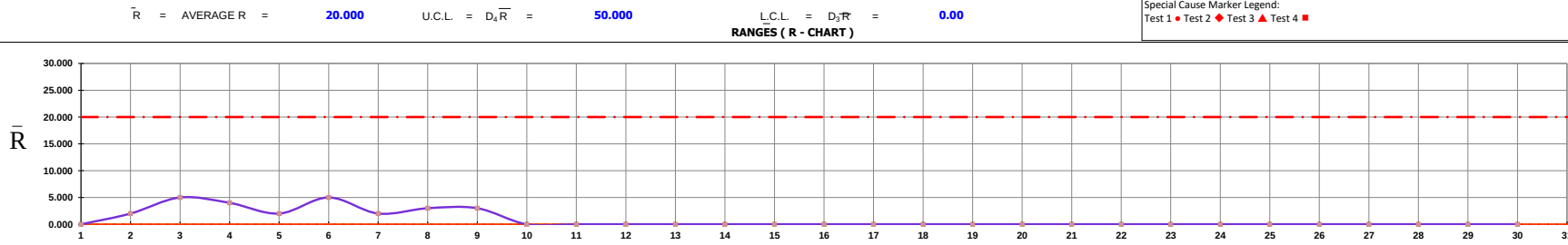
No. Dokumen	4-QAS-PP07-001
Revisi	00
Tgl Berlaku	2-Dec-16

MACHINE CT-1	SECTION QI	OPERATION CT-1	DATE CONTROL LIMITS CALC.		SPECIFICATION min 270 max 330		COURSE NO.	CONTROL ITEM (Δ) YES NO	☐
	MONTH Jun-25	CHARACTERISTIC Temperatur			SAMPLE SIZE / FREQUENCY n = 3		TYPE / SIZE : D2DSW2 / 0.50 PIC = DWI AGUS		

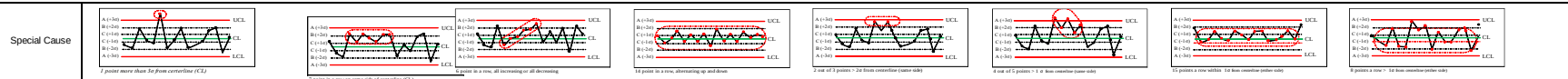


- | SPECIAL CAUSES | |
|----------------|--|
| ✓ | TEST 1: ANY POINT OUT OF CONTROL LIMIT |
| ✓ | TEST 2: 7 POINTS IN A ROW ON SAME SIDE OF CENTER LINE |
| ✓ | TEST 3: 6 POINTS IN A ROW ALL INCREASING OR DECREASING |
| ✓ | TEST 4: 14 POINTS IN A ROW, ALTERNATING UP & DOWN |
| ✓ | TEST 5: 2 OUT OF 3 POINTS > 2 STDEV FROM CENTER LINE (SAME SIDE) |
| ✓ | TEST 6: 4 OUT OF 5 POINTS > 1 STDEV FROM CENTER LINE (SAME SIDE) |
| ✓ | TEST 7: 15 POINTS IN A ROW WITHIN 1 STDEV OF CENTER LINE (EITHER SIDE) |
| ✓ | TEST 8: 8 POINTS IN A ROW > 1 STDEV FROM CENTER LINE (EITHER SIDE) |

ACTION INSTRUCTION	
1.	_____
2.	_____
3.	_____
4.	_____
5.	_____



SUBGROUP SIZE	A ₂	d ₂	D ₃	D ₄
2	1.881	1.128	0	3.267
3	1.023	1.693	0	2.574
4	0.729	2.059	0	2.281
5	0.577	2.326	0	2.115
6	0.483	2.534	0	2.004
7	0.4193	2.704	0.0756	1.9244
8	0.3726	2.847	0.1361	1.8639

[illegible]

CALCULATION	
$\bar{X}$	= 309.0
$\bar{R}$	= 3.3
S_{ST}	= 1.9
S_{LT}	= 4.934
THE PROCESS MUST BE IN CONTROL BEFORE CAPABILITY CAN BE DETERMINED	
C_{PK}	= $\min \left\{ \frac{\bar{X} - LSL}{3S_{ST}}, \frac{USL - \bar{X}}{3S_{ST}} \right\}$
	= $\min \left\{ \frac{5.91}{5.91}, \frac{2.77}{2.77} \right\}$
	= 2.77
C_p	= $\left[\frac{USL - LSL}{6S_{ST}} \right]$
	= $\left[\frac{4.34}{6.0} \right]$

	CHECKED BY	APPROVED BY
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